

Tableau Basics

Q1. Explain what Tableau is and why it is preferred over traditional spreadsheet tools for data analysis. Also mention any two industries where Tableau is commonly used and explain one use case.

Answer: Tableau is a leading data visualization and business intelligence tool that transforms raw data into interactive dashboards and insights. It is preferred over traditional spreadsheet tools like Excel because it handles large datasets efficiently, offers real-time connections, and provides intuitive drag-and-drop visualizations that make analysis faster and more accessible.

What Tableau Is

1. **Data Visualization & BI Platform:** Converts raw data into charts, maps, and dashboards.
2. **Drag-and-Drop Interface:** No coding required; users can build visuals by simply dragging fields.
3. **Wide Connectivity:** Connects to spreadsheets, SQL databases, cloud platforms, and APIs.
4. **Scalability:** Handles millions of rows of data smoothly, unlike spreadsheets that struggle at scale.
5. **Collaboration:** Dashboards can be shared via Tableau Server, Tableau Online, or Tableau Public.

Why Tableau Is Preferred Over Spreadsheets

1. **Data Size Handling:**
 - >> Tableau can handle **millions of rows** of data without slowing down.
 - >> Spreadsheets often **lag or crash** when datasets become very large.
2. **Visualization:**
 - >> Tableau provides **interactive, dynamic dashboards** with charts, maps, and filters.
 - >> Spreadsheets only offer **static charts** with limited interactivity.
3. **Real-Time Updates:**
 - >> Tableau supports **live connections** to databases and cloud sources, so data refreshes automatically.
 - >> Spreadsheets usually require manual refreshes or re-uploads to update data.
4. **Ease of Use:**
 - >> Tableau uses a **drag-and-drop interface** and even natural language queries.
 - >> Spreadsheets depend on **formulas and manual setup**, which can be complex and error-prone.

5. Collaboration:

- >> Tableau dashboards can be shared via **cloud or server** with permissions and version control.
- >> Spreadsheets rely on **file sharing**, which often leads to version conflicts.

Industries Where Tableau Is Commonly Used

1. Retail & E-Commerce

Use Case: Analyzing customer purchase behavior.

Example: A retailer uses Tableau to track sales by region and product category, helping optimize inventory and marketing campaigns.

2. Healthcare

Use Case: Patient outcome and resource tracking.

Example: Hospitals use Tableau dashboards to monitor patient admissions, treatment effectiveness, and resource utilization, improving efficiency and care quality.

Q2. After connecting to the Sample – Superstore dataset:

- List the main types of fields available (examples: time-based, geographical, numerical).
- Identify any four Dimensions and any four Measures from the dataset.

Answer:

a) Main Types of Fields Available

When you connect to the Sample – Superstore dataset in Tableau, you'll find three broad categories of fields:

- >> **Time-based fields** - e.g., Order Date, Ship Date
- >> **Geographical fields** - e.g., Country, State, City, Postal Code
- >> **Numerical fields** - e.g., Sales, Profit, Quantity, Discount

b) Examples of Dimensions (Blue fields)

Dimensions are categorical fields used to segment and describe data. Four examples from the dataset are:

1. Category
2. Sub-Category
3. Region
4. Customer Name

Examples of Measures (Green fields)

Measures are numerical fields that can be aggregated (sum, average, etc.). Four examples from the dataset are:

1. Sales
2. Profit
3. Quantity
4. Discount